> Sometimes by the help of a sunbeam, and sometimes by the convenience of a comet, [they] glided from sphere to sphere, as a bird hops from bough to bough. In a very little time [they] posted through the Milky Way …

We are, even now, discovering vast numbers of gas and dust disks around young stars—the very structures out of which, in our solar system four and a half billion years ago, the Earth and the other planets formed. We're beginning to understand how fine dust grains slowly grow into worlds; how big Earthlike planets accrete and then quickly capture hydrogen and helium to become the hidden cores of gas giants; and how small terrestrial planets remain comparatively bare of atmosphere. We are reconstructing the histories of worlds—how mainly ices and organics collected together in the chilly outskirts of the early Solar System, and mainly rock and metal in the inner regions warmed by the young Sun. We have begun to recognize the dominant role of early collisions in knocking worlds over, gouging huge craters and basins in their surfaces and interiors, spinning them up, making and obliterating moons, creating rings, carrying, it may be, whole oceans down from the skies, and then depositing a veneer of organic matter as the neat finishing touch in the creation of worlds. We are beginning to apply this knowledge to other systems.

In the next few decades we have a real chance of examining the layout and something of the composition of many other mature planetary systems around nearby stars. We will begin to know which aspects of our system are the rule and which the exception. What is more common—planets like Jupiter, planets like Neptune, or planets like Earth? Or do all other systems have Jupiters and Neptunes and Earths? What other categories of worlds are there, currently unknown to us? Are all solar systems embedded in a vast spherical cloud of comets? Most stars in the sky are not solitary suns like our own, but double or multiple systems in which the stars are in mutual orbit. Are there planets in such systems? If so, what are they like? If, as we now think, planetary systems are a routine consequence of the origin of suns, have they followed very different evolutionary paths elsewhere? What do elderly planetary systems,